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Effective Reduction of Wavenumbers of Rotational and Vibration–Rotational Transitions of LiH $X\,^1\Sigma^+$

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A deformationally self-consistent procedure is applied to reduce the wavenumbers of 594 pure rotational and vibration–rotational transitions of ^{7}Li^{1}H, ^{6}Li^{1}H, ^{7}Li^{2}H, and ^{6}Li^{2}H to 15 unconstrained and 2 constrained parameters with $\hat{\sigma} = 1.098$, $\sigma = 0.000554$ cm^{-1}, and $F = 7.704 \times 10^{14}$. The quality of the Born–Oppenheimer approximation and the extent of its breakdown in LiH are reexamined. © 1997 Academic Press

INTRODUCTION

The availability of four isotopic variants and well-resolved infrared and microwave spectra make LiH a suitable molecule for discussion of a breakdown in the Born–Oppenheimer (BO) approximation and examination of the status of theoretical models contemporarily used in spectral analysis. Precise spectral data of LiH $X\,^1\Sigma^+$, obtained by the application of Fourier-transform spectrometers and diode lasers, comprise pure rotational transitions in the millimeter-wave (*1*) and far-infrared (*2–4*) regions, and vibration–rotational transitions in the mid-infrared region (*2, 5*), supplemented by accurate measurements by Matsushima *et al.* (*6*), and Bellini *et al.* (*7*).

The first analysis of the LiH spectra, performed by Maki *et al.* (*2*), failed to reproduce them satisfactorily. Because the differences between the theoretical and experimental results were especially pronounced for highly excited rotational states (*2*), two sources of problems have been suggested:

(i) nonadiabatic effects that might be large for distant rotational states; in such circumstances the perturbation calculation of the interaction between electrons and nuclei could be inadequate up to second order (*8*);

(ii) centrifugal distortion resulting in change of the BO reference conformation of molecules (*9–11*), being especially large for soft molecules and highly excited rotational states.

The inclusion of nonadiabatic effects in a treatment of the vibration–rotational spectra of LiH has allowed their successful analysis by Ogilvie (*12, 13*). However, the procedures applied by him have omitted the important contribution of the rotational g-factor g_J and dipole moment, which are hidden in the radial function (*15*)

$$\alpha(R) = m_e \sum_{i=0} \sum_{\mu=a,b} t_i^{\mu} m_{\mu}^{-1} z^i, \qquad [1]$$

describing nonadiabatic rotational effects reflecting the fact that electrons fail to follow perfectly the nuclei rotating about the center of molecular mass. In Eq. [1] $z = 2(R - R_0)/(R + R_0)$ denotes Ogilvie's variable for reduced displacement from equilibrium R_0 (*16*); m_e, m_a, and m_b are masses of electron and nuclei; and R_0 is the equilibrium internuclear distance according to the BO procedure for $J = 0$. To overcome this difficulty, Tiemann and Ogilvie (*8*) fitted then available transitions of LiH using t_0^{Li} and t_0^{H} constrained to values consistent with their relations,

$$t_0^{Li} = m[g_0/m_p + 2\mu_0/(eR_0m_H)], \qquad [2]$$

$$t_0^{H} = m[g_0/m_p - 2\mu_0/(eR_0m_{Li})], \qquad [3]$$

to known experimental values of electric dipolar moment μ_0 and rotational g-factor g_0, both measurable in states of small vibrational v and rotational J quantum numbers (*17, 18*), respectively. In the above equations m and m_p are reduced mass of LiH and protonic mass, respectively; e is electric charge.

Correcting errors in the previous paper (*16*), Ogilvie (*19*) applied his approach to reduce the most extensive set of 594 pure rotational and vibration–rotational transitions to 18 unconstrained and 2 constrained parameters

$$t_0^{Li} = 0.75335, \quad t_0^{H} = -0.76823, \qquad [4]$$

calculated according to Eqs. [2] and [3], respectively.

Because of the simple structure of LiH, electronic calculations of the rotational g-factor $g_J(R)$ and the electric dipolar moment $\mu_J(R)$ as a function of internuclear distance have

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been performed (*15*). From those data and the relationships (*15*)

$$\sum_{i=0} t_i^{\mathrm{Li}} m_{\mathrm{Li}}^{-1} z^i = m[g_J(R)/m_{\mathrm{p}} + 2\mu_J(R)/(eRm_{\mathrm{H}})], \quad [5]$$

$$\sum_{i=0} t_i^{\mathrm{H}} m_{\mathrm{H}}^{-1} z^i = m[g_J(R)/m_{\mathrm{p}} - 2\mu_J(R)/(eRm_{\mathrm{Li}})], \quad [6]$$

Ogilvie *et al.* evaluated parameters $t_i^{\mathrm{Li,H}}$ and then reproduced 557 transitions of LiH with all $t_i^{\mathrm{Li,H}}$ constrained. This approach for the first time allowed evaluation of nonadiabatic vibrational effects of high order, due to vibrational inertia of electrons, described according to the radial function

$$\beta(R) = m_{\mathrm{e}} \sum_{i=0} \sum_{\mu=a,b} s_i^{\mu} m_{\mu}^{-1} z^i. \quad [7]$$

The evaluation of these effects from only data of frequency type was otherwise impossible.

A second explanation of the problem reported by Maki *et al.* (*2*) was that they arise due to a deformational action of the centrifugal force acting in all systems with rotational degrees of freedom (*11*). Exploiting this idea a new method of investigating molecular spectra, based on a model of a deformable body (MDB) and implemented in a deformationally self-consistent (DSC) procedure (*9–11*), was developed. It was first applied in a simple analysis of vibration–rotational spectra of single isotopic variants of some diatomic molecules (including LiH) (*9, 10*), and then in a quantitative analysis of adiabatic and nonadiabatic effects in vibration–rotational spectra of GaH in four isotopic variants (*11*) providing a satisfactory reproduction of the experimental data using only 18 parameters compared to 23 in a previous approach (*14*).

The main objective of this work is to apply the DSC procedure to quantitative analysis of adiabatic and nonadiabatic effects in the pure rotational and vibration–rotational spectra of LiH $X\,^1\Sigma^+$, and to compare the method based on the DBM with the conventional approach. We discuss also the constraints used by Ogilvie (*19*), suggesting small modifications of the formulas employed, permitting work in the scheme with dependent and unconstrained parameters $t_0^{\mathrm{Li,H}}$.

TREATMENT OF SPECTRAL DATA

To conduct an analysis of LiH spectra we adopt a procedure similar to that for GaH (*11*). The point of departure is a wave equation of Herman and Asgharian (*20*)

$$\left\{ -\frac{\hbar^2}{2m}\frac{d}{dR}[1 + \beta(R)]\frac{d}{dR} + \mathrm{U}_J(R) - E_{vJ} \right\} \psi_{vJ}(R) = 0, \quad [8]$$

$$U_J(R) = \frac{\hbar^2 J(J+1)[1 + \alpha(R)]}{2mR^2} + \mathrm{V}(R) + V'(R) \quad [9]$$

$$V(R) = c_0 z^2 (1 + \sum_{i=1} c_i z^i), \quad [10]$$

$$V'(R) = m_{\mathrm{e}} \sum_{i=1} \sum_{\mu=a,b} u_i^{\mu} m_{\mu}^{-1} z^i, \quad [11]$$

in which $V(R)$ describes a potential energy independent of mass and $V'(R)$ is an adiabatic term reflecting the dependence of internuclear potential energy, on not only the distance between nuclei but also their relative momenta, whereas $\alpha(R)$ and $\beta(R)$ take into account nonadiabatic rotational and vibrational effects.

In the first step we expand $U_J(R)$

$$U_J(\eta) = \sum_{n=0} b_n^J \eta^n, \quad [12]$$

into a series of a Dunham variable dependent on J,

$$\eta = (R - R_J^{\mathrm{ad}})/R_J^{\mathrm{ad}}, \quad [13]$$

recognizing that vibrational displacements of nuclei in the vibration–rotational diatomic systems take place in the vicinity of the dynamical equilibrium conformation R_J^{ad}, dependent on J (*11*), and not the BO conformation R_0. The R_J^{ad} appearing in [13] are defined according to a criterion for a minimum

$$\left(\frac{dU_J(R)}{dR}\right)_{R_J^{\mathrm{ad}}} = 0, \quad [14]$$

allowing the calculation of a modified reference conformation for each rotational state J, and the removal of the linear term $b_1^J \eta$ in the expansion [12]. Accordingly, we achieve an effective wave equation

$$\left[-B_J \frac{d^2}{d\eta^2} + a_0^J \eta^2 (1 + \sum_{s=1} a_s^J \eta^s) - (E_{vJ} - b_0^J) \right] \psi_{vJ}(\eta) = 0, \quad [15]$$

in which

$$a_0^J = b_2^J, \quad a_{s>0}^J = b_{s+2}^J / a_0^J, \quad B_J = \hbar^2/[2m(R_J^{\mathrm{ad}})^2], \quad [16]$$

which is amenable to straightforward solution according to Dunham's quasi-classical scheme (*11*) generating eigenvalues

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TABLE 1
Mass-Independent Radial Parameters[a] of LiH $X\,^1\Sigma^+$

i	c_i	t_i^H	u_i^{Li} cm^{-1}	u_i^H cm^{-1}
0	65724.23(11) cm^{-1}	[−0.76823]		
1	−0.8972356(81)	1.5053(25)	−57216(141)	−51877.8(86)
2	0.348310(21)	−2.152(39)	1259489(1342)	32869(98)
3	−0.088435(87)	2.646(71)	−129285(6227)	−6113(646)
4	−0.04517(24)			
R_0	1.59491035(16)Å			
t_0^{Li}	[0.75335]			

[a] The normalized standard deviation of the fit of 594 data was $\hat{\sigma} = 1.098$, standard deviation $\sigma = 0.000554$ cm^{-1} and F-value 7.704×10^{14}. Values enclosed in brackets are constrained.

$$E_{vJ} = b_0^J + \sum_{k=0} Y_{k0}(v + 1/2)^k. \qquad [17]$$

The coefficients Y_{k0} appearing in [17] are purely vibrational coefficients (*21*) in which substitutions $\{R_0, a_n\} \rightarrow \{R_J^{ad}, a_n^J\}$ can be made. As eigenvalues [17] contain unknown parameters R_J^{ad} defined according to the general condition for equilibrium [14] we can effect quantitative analysis of spectral data working in an iterative scheme, called the DSC procedure, described in detail elsewhere (*11*).

CONSTRAINED AND DEPENDENT PARAMETERS

Detailed analysis of constraints [4] applied by Tiemann and Ogilvie (*8*), and by Ogilvie (*19*), indicates that they are calculated for[1]

$$g_0^{7,1} \approx g_{J=1}^{7,1} = -0.65842(17),$$
$$g_0^{7,2} \approx g_{J=1}^{7,2} = -0.27674(11) \qquad [18]$$
$$\mu_0 = 1.94407(22) \times 10^{-29}\ \text{C m} \qquad [19]$$

by arithmetic averaging of Eqs. [2] and [3]

$$t_0^{Li} = \{m^{7,1}[g_0^{7,1}/m_p + 2\mu_0/(eR_0m_1)] + m^{7,2}[g_0^{7,2}/m_p + 2\mu_0/(eR_0m_2)]\}/2, \qquad [20]$$

$$t_0^{H} = \{m^{7,1}[g_0^{7,1}/m_p - 2\mu_0/(eR_0m_7)] + m^{7,2}[g_0^{7,2}/m_p - 2\mu_0/(eR_0m_7)]\}/2, \qquad [21]$$

[1] The value $\mu_0 = 1.94407(22) \times 10^{-29}$ C m is the equilibrium value of the LiH dipole moment obtained by extrapolating to $v = -1/2$ the experimental values of μ for ^{7}Li^{1}H, measured in the vibrational state $v = 0, 1, 2$ (J. F. Ogilvie, private communication).

in which $m^{k,l}$ denotes reduced mass and m_k is a mass of the nucleus, whereas $g_J^{k,l}$ is a rotational g-factor, for a given isotopic variant ^{k}Li^{l}H.

Inspection of these equations reveals some inconsistency in the method of determination of $t_0^{Li,H}$ producing values in [4]. Namely, Eqs. [20] and [21] contain R_0 that appears also in a mass-independent parameter $U_{0,1} = \hbar^2/(2R_0^2)$ used by Tiemann and Ogilvie (*8*), and by Ogilvie (*19*), as a fitted parameter. Hence, instead of constraints according to (*4*), the R_0-dependent parameters $t_0^{Li,H}$, given by Eqs. [20] and [21], should be applied in calculations. The same quantity cannot be simultaneously set up to the constant value and fitted as a free parameter.

An additional approximation used by Tiemann and Ogilvie (*8*) and by Ogilvie (*19*) is $g_0 \approx g_{J=1}$. As has been shown in our previous work (*11*), such estimation is charged with some error due to the rotational dependence of g_J in the *vibrationless* state $v = 0$, $J > 0$

TABLE 2
Mass-Dependent Deformational Parameters $x_J = (R_J^{ad} - R_0)/R_0$

J	^{7}Li^{1}H	^{7}Li^{2}H	^{6}Li^{1}H	^{6}Li^{2}H
0	0.0002488663	0.0001415245	0.0002545259	0.0001471853
1	0.0004775239	0.0002703396	0.0004879609	0.0002807803
2	0.0009347916	0.0005279547	0.0009547815	0.0005479540
3	0.0016205754	0.0009143390	0.0016548897	0.0009486737
4	0.0025347359	0.0014294487	0.0025881403	0.0014828910
5	0.0036770908	0.0020732232	0.0037543435	0.0021505427
10	0.0128028068	0.0072190629	0.0130703187	0.0074870769
15	0.0275881412	0.0155656055	0.0281631878	0.0161424053
20	0.0479900735	0.0270945265	0.0489888063	0.0280970638
25	0.0739832087	0.0417871894		0.0433315797
31		0.0635765084		

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TABLE 3
Origins of Bands[a] of $^7Li^1H$ and $^7Li^2H$ Determined[b] from LiH Spectral Data and Calculated from Radial Parameters in Table 1

$^7Li^1H$			$^7Li^2H$		
$v' \rightarrow v''$	E_{Det}	E_{Cal}	$v' \rightarrow v''$	E_{Det}	E_{Cal}
$0 \rightarrow 1$	1359.708186(89)	1359.708257	$0 \rightarrow 1$	1029.06395(15)	1029.06421
$1 \rightarrow 2$	1314.85162(31)	1314.85132	$1 \rightarrow 2$	1003.59800(13)	1003.59819
$2 \rightarrow 3$	1270.91362(97)	1270.91073	$2 \rightarrow 3$	978.53393(47)	978.53423
$3 \rightarrow 4$[c]		1227.81688	$3 \rightarrow 4$		953.85027

[a] In cm^{-1}.
[b] Origins of bands were determined from LiH spectral data using Crout's programme.
[c] Accurate values of origins of bands $3 \rightarrow 4$ are indeterminate from available spectral data.

$$g_J(R_J^{ad}) = m_p \sum_{i=0} \sum_{\mu=a,b} t_i^\mu m_\mu^{-1} z_J^i, \quad [22]$$

where $z_J = 2(R_J^{ad} - R_0)/(R_J^{ad} + R_0)$ is Ogilvie's variable z in rotational state J. Consequently, constant terms of the expansions [22] are

$$t_0^a = m[G_J/m_p + 2\mu_0/(eR_0m_b)], \quad [23]$$

$$t_0^b = m[G_J/m_p - 2\mu_0/(eR_0m_a)], \quad [24]$$

in which

$$G_J = g_J(R_J^{ad}) - m_p \sum_{i=1} \sum_{\mu=a,b} t_i^\mu m_\mu^{-1} z_J^i. \quad [25]$$

Bearing in mind Eqs. [20] and [21] we find the final form of the dependent parameters $t_0^{a(b)}$

$$t_0^{Li} = \{m^{7,1}[G_J^{7,1}/m_p + 2\mu_0/(eR_0m_1)] + m^{7,2}[G_J^{7,2}/m_p + 2\mu_0/(eR_0m_2)]\}/2 \quad [26]$$

$$t_0^{H} = \{m^{7,1}[G_J^{7,1}/m_p - 2\mu_0/(eR_0m_7)] + m^{7,2}[G_J^{7,2}/m_p - 2\mu_0/(eR_0m_7)]\}/2, \quad [27]$$

which we suggest to be used in calculations.

APPLICATIONS

In this paper we apply the DSC procedure to evaluate radial parameters of LiH in four isotopic variants, working in a scheme with constrained and dependent parameters $t_0^{a(b)}$. As input data we used 594 pure rotational and vibration–rotational transitions of $^7Li^1H$, $^6Li^1H$, $^7Li^2H$, and $^6Li^2H$ reported in (*1–7*). As the starting wave equation [8] and the

TABLE 4
Origins of Bands[a] of $^6Li^1H$ and $^6Li^2H$ Determined[b] from LiH Spectral Data and Calculated from Radial Parameters in Table 1

$^6Li^1H$			$^6Li^2H$		
$v' \rightarrow v''$	E_{Det}	E_{Cal}	$v' \rightarrow v''$	E_{Det}	E_{Cal}
$0 \rightarrow 1$	1373.36577(35)	1373.36578	$0 \rightarrow 1$	1047.50412(24)	1047.50429
$1 \rightarrow 2$	1327.58381(63)	1327.58308	$1 \rightarrow 2$	1021.10463(32)	1021.10460
$2 \rightarrow 3$[c]		1282.74954	$2 \rightarrow 3$	995.15382(760)	995.12867

[a] In cm^{-1}.
[b] Origins of bands were determined from LiH spectral data using Crout's programme.
[c] An accurate value of the origin of band $2 \rightarrow 3$ of $^6Li^1H$ is indeterminate from available spectral data.

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input data set are the same in DSC and standard approach, a direct comparison of both approaches is possible.

Table 1 presents the radial parameters obtained in the DSC scheme with two parameters t_0^{Li} and t_0^{H} constrained to values consistent with their relations to the known dipolar moment and rotational g-factor (*19*); all other parameters were constrained to zero. The uncertainty in parentheses is one estimated standard deviation in units of the last quoted digit of the fitted values.

Fitted parameters in various sets were tested according to the following criteria:

(i) minimum number of fitted parameters N;

(ii) optimum value of the normalized standard deviation $\hat{\sigma} \approx 1$;

(iii) maximum value of the F statistic (*18*);

(iv) optimum values of standard errors σ_i of fitted parameters and correlation coefficients $\mathrm{cc}(i, j)$.

In calculations we used the vibrational Dunham coefficients

$$Y_{k0} = \sum_{h=0}^{4} Y_{k0}^{2h}, \qquad [28]$$

up to sixth order for $0 \leqslant k \leqslant 4$, including a_n, $0 \leqslant n \leqslant 6$, potential parameters. Consequently, the expansion for the effective potential energy [12] contains derivatives up to eighth order. The required Y_{k0}^{2h} terms were derived using a Maple processor (*22*).

The proposed iterative DSC procedure is rapidly convergent and in the case of LiH requires 10 iterations to obtain parameters in a final set such that values alter less than their standard errors, and affect insignificantly the precision of the calculations. Initial values of fitted parameters were taken from Ogilvie's calculations (*19*).

In Table 2 parameters $x_J = (R_J^{\mathrm{ad}} - R_0)/R_0$ for LiH in four isotopic variants are reported; they are evaluated with an accuracy $\epsilon = 10^{-20}$ to be achieved in 60 bisections. The parameter $x_1 = (R_1^{\mathrm{ad}} - R_0)/R_0$, calculated for $J = 1$, may be considered an indicator of molecular susceptibility to centrifugal distortion, or in other words, an indicator of molecular rigidity.

The quantity R_0^{ad}, appearing in Table 2 for $J = 0$, stands for the adiabatic reference conformation defined according to a criterion of the minimum of potential energy $U_0(R) = V(R) + V'(R)$.

Tables 3 and 4 report the origins of bands of LiH in four isotopic variants calculated with the radial parameters from Table 1, and from fitting the experimental wavenumbers of the observed transitions in each band separately.

RESULTS AND DISCUSSION

Detailed analysis of the results obtained indicates that application of this DSC procedure enables reduction of wavenumbers of 594 rotational and vibration–rotational transitions of LiH to 15 radial parameters rather than 18 in the standard approach (*19*). The calculation made in a scheme with dependent parameters $t_0^{a(b)}$ given by [23] and [24] instead of constraints [4] yielded a set of fitted parameters identical to that reported in Table 1; according to this approach the evaluated parameters $t_0^{a(b)}$ took the values

$$t_0^{\mathrm{Li}} = 0.75288(30), \quad t_0^{\mathrm{H}} = -0.76870(13), \qquad [29]$$

slightly different from the ones used by Ogilvie (*19*).

In preliminary fits of the data, we evaluated $c_5 = -0.0047(45)$ and $c_6 = 0.0230(78)$ with $\hat{\delta} = 1.078$ and $F = 6.789 \times 10^{14}$; after removing c_6 from the set of fitted parameters we obtained $c_5 = 0.0080(19)$ and $\hat{\delta} = 1.083$, $F = 7.333 \times 10^{14}$. Additionally we discovered that in the absence of both c_5 and c_6 we obtain $\hat{\sigma} = 1.098$ and $F = 7.704 \times 10^{14}$. Hence by virtue of a criterion of maximum F statistic we deleted them from the parameters in the final sets. The omission of c_6 and c_5 causes only slightly deteriorated accuracy of reproduction of the spectral data.

The above results differ from those of Ogilvie (*19*), who evaluated $c_5 = -0.0222(63)$ and $c_6 = 0.0397(95)$ with $\hat{\sigma} = 1.12$ and $F = 2.20 \times 10^{15}$, and they were indispensable in fitting the LiH data.

Worth mentioning is the absence of t_1^{Li}, u_4^{H}, and u_5^{H}; instead parameters t_3^{H} and u_3^{Li} appear. In the previous work $t_1^{\mathrm{Li}} = 0.105(56)$ (*19*) is charged with error comparable to the value of the parameter; a similar condition applies for $u_5^{\mathrm{H}} = -12.6(2.7) \times 10^4\ \mathrm{cm}^{-1}$ which is also poorly evaluated. We conclude that nonadiabatic rotational effects are significant for only hydrogen nuclei, whereas for nuclei of lithium they are negligible. In the case of adiabatic effects they are uniformly distributed on the two nuclear centers without domination of the hydrogen nucleus.

Comparison of values of the determined reference conformation R_0 indicates that the DSC approach produces a more precise (in one order of magnitude) $R_0 = 1.59491035(16)$ Å than the standard method yielding $R_0 = 1.5949107(14)$ Å.[2] The latter, like c_1, u_1^{Li}, u_1^{H}, and u_2^{H}, is identical (within the quoted error) in both approaches.

A parameter $x_1 = (R_1^{\mathrm{ad}} - R_0)/R_0 \approx 10^{-4}$ is a source of important information on susceptibility of LiH to centrifugal distortion. The calculations performed for GaH gave (*11*) $x_1 \approx 10^{-4}$–10^{-5}, whereas for BrCl and GeS (*23*) $x_1 \approx 10^{-7}$. In view of this, GaH and LiH may be viewed as *semirigid*, whereas BrCl and GeS may be viewed as *rigid* molecules. It has been shown that in the former case we need in the DSC approach three to five parameters less than in the standard

[2] This value has been calculated from a fitted $U_{0,1} = \hbar^2/(2R_0^2)$ parameter with uncertainty estimated according to the error propagation rule, including the errors in fundamental constants (J. F. Ogilvie, private communication).

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approach, whereas for the latter we require one parameter less to obtain comparable accuracy of reproduction of spectral data (*23*).

The absence of c_5, c_6, and u_5^{H} in the set of DSC parameters indicates that in the standard approach they play a role of *compensating* parameters. Hence the presence of c_5, c_6, and u_5^{H} in the fit ensures a good statistical characteristic, but these parameters lack physical meaning. In the DSC approach their presence in the fit is superfluous.

Despite the absence of these parameters the remaining ones satisfactorily reproduce the origins of bands of four isotopomers of LiH (see Tables 3 and 4). Thus the present approach produces physically and chemically meaningful radial parameters, and reduction of 594 spectral data of LiH in four isotopic variants to 15 parameters is more efficient than that obtained previously (*19*).

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